JalSense · build & demo guide

Smart RO bypass: reads TDS and clarity, then runs the bypass pump or the RO pump by itself.

`sketch: JalSense_auto.ino` `ESP32 Dev Module`

HOW IT DECIDES (CHECKED IN THIS ORDER)

1. Lock mode on? → **RO**
2. Sensor fault (TDS < 150 mV, turbidity < 0.3 V)? → **RO** (fail-safe)
3. Clarity < **85 %** of clear water? → **RO**
4. TDS ≥ **300 ppm** ? → **RO**
5. Otherwise → **BYPASS** (pump A, through the flow sensor)

Each dispense is 250 mL. Auto mode repeats it by itself (3 s rest), up to 1 L, then switches off.

WIRING THAT WORKS

Part	Connects to
TDS meter (analog)	+ → 3V3, − → GND, A → **GPIO34**
Turbidity module	VCC → 5 V, GND, A → R1 → node → **GPIO35**; R2 node → GND
DS18B20 probe	red 3V3, black GND, yellow **GPIO4**; R5 from data to 3V3
Flow sensor YF-S201	red 5 V, black GND, yellow → R3 → node → **GPIO27**; R4a + R4b node → GND
OLED (SH1106 chip)	VCC 3V3, GND, SDA **GPIO21**, SCL **GPIO22**
Relay IN1 / IN2	**GPIO25** (pump A, bypass) / **GPIO26** (pump B, RO)
Relay power	VCC → **5 V**, GND; **JD-VCC jumper stays ON**
Relay output	COM1, COM2 → 5 V; pump + → **NO**; pump − → GND
Lock / dispense	GPIO23 to GND = lock; BOOT button = dispense

Resistors R1, R2, R3, R4a, R4b, R5 are all 5.5 kΩ. Divider ratios are unchanged, so no code change.

Power: ESP32 from **USB**; pumps and relay from the **5 V 2 A adapter**. Join the grounds. Never feed the ESP32's 5V pin from the adapter while USB is plugged in. Put the 470 µF capacitor across 5 V and GND.

UPLOAD THE CODE

1. Arduino IDE: board **ESP32 Dev Module**, **Upload Speed 115200**.
2. Libraries: OneWire, DallasTemperature, Adafruit SSD1306, Adafruit GFX.
3. If it stalls at "Connecting...": unplug the adapter, hold **BOOT**, use a short USB cable.

CALIBRATE ONCE (SERIAL MONITOR, 115200)

`p<ppm>`	probe in a 400–700 ppm solution, e.g. `p550` (match your pen)
`c`	probe in clear water → saves the 100 % clarity reference
`a` / `b`	10-second pump run; then `ma<mL>` / `mb<mL>` with the water collected
`s`	prints every reading; `o` auto on/off; `x` resets calibration

BEFORE EVERY DEMO (2 MINUTES)

- Tank filled, both pumps **fully underwater**, tubing short (30–40 cm) and not kinked.
- Probes in the water: TDS, turbidity, temperature. Wipe the turbidity gap clean.
- Type `s` : TDS and clarity look right; temperature is real (not "probe missing").
- Join Wi-Fi **JalSense-Demo** (password `jalsense123`), mobile data off, open `http://192.168.4.1` .

DEMO SCRIPT (ABOUT 3 MINUTES)

Step	Do this	Show
1 Clean water	Under 300 ppm, clear. Press BOOT, or let auto mode start.	BYPASS, pump A, flow count
2 Salty water	Add a pinch of salt (about 0.4 g per litre). TDS climbs past 300.	Flips to RO, pump B
3 Cloudy water	A few drops of milk. Clarity falls below 85 %.	RO: "cloudy water"
4 Probe out	Lift the TDS probe out of the water.	Fault → RO (fail-safe)
5 Lock + phone	Turn on Lock on the phone page, then off.	LOCK → always RO

Counters: **Saved** grows after a bypass dispense (reject avoided); **Reuse** grows after an RO dispense. Both are estimates at a 75 % reject share, and reset at power-up.

IF SOMETHING GOES WRONG

OLED garbage	Chip is SH1106: keep `OLED_IS_SH1106 = true` . Reseat SDA and SCL.
Pump silent	Click on `a` ? No click: relay VCC and JD-VCC on 5 V, jumper on. Click: check COM on 5 V and red lead on NO.
Both pumps run	Pump red leads belong on **NO**, never on COM or NC.
"No flow" stop	Pipe too long or too high, or tank empty. Flow sensor needs about 1 L/min.
Upload fails	Speed 115200, adapter unplugged, hold BOOT, short cable.
TDS looks wrong	Pin mV near 2.3 V means over range. Send `x` , then recalibrate with `p` .

BE HONEST ABOUT THE LIMITS

- TDS and clarity say nothing about **bacteria**, **arsenic** or **fluoride**. UV and UF stay on in every path, and **Lock mode** forces RO where those are a concern.
- The analog TDS meter reads **0–1000 ppm**. Water far above that can read wrongly low.
- Hobby sensors drift: recalibrate against a TDS pen before each demo.
- This is a prototype. It has not been tested for drinking-water safety.

Files: JalSense_auto.ino · circuit and breadboard diagrams (v2) · working-model video.